

ECON1010

Introductory Microeconomics

Tutorial 12 | Economics of Information



This Week in ECON1010

CML 6 is due **Friday** 4.00pm May 29
(FINAL ONE!!)

Tutor evaluations are still open!!!



Assessment

85% **7**

75% **6**

65% **5**

50% **4**

47% **3**

30% **2**

0% **1**

CMLs (25%)

- Computer Managed Learning
- Each worth 5%
- Best 5/6 count towards final grade

CLEAR-JE (20%)

- Contextualised Learning of Economics via Authentic Reflection using Journal Entries
- Each worth 10%
- Best 2/3 count towards final grade

Final Exam (55%)

Tools and Resources

I need HELP...

1. **Ed Discussion Board**
(Blackboard)
2. **Consultation** (8.00am
Mon, 39-125A)
3. **Staff email**
econ1010@uq.edu.au

I REALLY need HELP...

- <https://my.uq.edu.au/contact/student-central>

ECON1010 Consult Timetable

Semester 1 2026

Time	Mon	Tue	Wed	Thu	Fri
8:00-8:30	CONSULT - SAM 39-125A				
8:30-9:00					
9:00-9:30					
9:30-10:00					
10:00-10:30				CONSULT - LUCAS 39-125A	
10:30-11:00					
11:00-11:30	CONSULT - GABE 39-125A	CONSULT - BORIS 39-125A	CONSULT - IKMAL 39-125A	CONSULT - NIK 39-125A	CONSULT - CARL 39-641
11:30-12:00					
12:00-12:30					
12:30-13:00					
13:00-13:30					LECTURE
13:30-14:00		CONSULT - DYLAN 39-125A			
14:00-14:30				CONSULT - GUANCHU 39-125A	
14:30-15:00					
15:00-15:30					
15:30-16:00					
16:00-16:30			CONSULT - KIRTAN 39-125A	CONSULT - NANDAN 39-125A	
16:30-17:00					
17:00-17:30					

Student Central

Need help? We've got you covered.

Contact us

Monday to Friday

- ✉ Email
- ☎ 1300 275 870 (8.30am-5pm)
- 📍 Building 42, St Lucia
- 📍 Building 8101A, Gatton

Chat – 1:51 (1 Minute) wait

Emergency help

For immediate risk:

- UQ Campus Security
- ☎ 07 3365 3333 (24/7)

Off-campus emergency

- ☎ 000 (24/7)

Crisis support

For urgent mental health support:

- UQ Counselling and Crisis Line
- ☎ 1300 851 998 (24/7)

Text a Crisis Counsellor

- ☎ 0488 884 115 (4.30pm-8am)

Sexual Misconduct Support Unit (SMSU)

Tutorial Slide Access



<https://github.com/uqstay25/ECON1010/tree/main>

- Use the GitHub link to access all tutorial PowerPoints
- Included is a revision slide deck.
- USE VERY CAUTIOUSLY
- This doc is by no means exhaustive, and AI was used to create it so it may be unreliable!!!
- I am not saying this has everything you need for the exam
- Read README.md

Economics of Information

Question 1a

In your own words, what do you understand by:

- i) Principal–Agent problem
- ii) Adverse Selection problem
- iii) Moral Hazard problem
- iv) Costly to Fake principle



Question 1a

i) Principal–Agent problem

Both principal and agent want to work together to maximise profits, but the agent's profit comes at the principal's cost. The principal is particularly vulnerable as the agent holds more knowledge and expertise over the principal's property and the market (shares, rental house, contract for sale etc).

The principal–agent problem occurs when an agent hired to act on behalf of a principal has different incentives and more information than the principal, allowing them to act in their own interest instead. For example, shareholders (principals) hire a CEO (agent) to run a company, but the CEO may focus on boosting short-term profits or expanding the company for personal bonuses and prestige rather than maximising long-term shareholder value.

ii) Adverse Selection problem

When there are consumers (or products) in a market with hidden weaknesses (defects) such as a risky person taking out insurance (faulty second-hand car), the market will naturally appeal to these risky consumers (bad products) as they have the highest benefit in a transaction. The increase in risky consumers (bad products) will result in increased prices (lowered average quality).

Adverse selection occurs when one party in a transaction has more information than the other before the deal is made, causing lower-quality participants or products to dominate the market. For example, in the used car market, sellers know whether a car is reliable or a “lemon,” but buyers do not. Because buyers cannot tell the difference, they offer only an average price, leading owners of good cars to leave the market while poor-quality cars remain.

Question 1a

iii) Moral Hazard problem

Once people are in an agreement, they tend to change their behaviour to maximise their benefits. When offered assurance or protection, the natural incentive to be careful disappears.

Example: If you have car-insurance, you'll drive more recklessly. This results in more crashes overall and then higher premiums for the whole market.

Moral hazard occurs when a person or firm takes greater risks because they do not bear the full consequences of their actions after an agreement is made. For example, someone with comprehensive car insurance may drive less carefully because the insurance company will cover most of the repair costs if they crash. The problem arises because protection from risk changes behaviour.

Question 1a

iv) Costly to Fake principle

The Costly to Fake principle is where a supplier (consumer) can show to a consumer (supplier) a piece of information, or signal, to verify the value of their product (self) – which cannot be easily be faked.

Example 1:

Used Car Market Supplier: “I have a good value car (information). It will not break down on you.”

Consumer: “I’m not sure I can believe you. Where is the proof?”

Supplier: “I will offer you a warranty (signal). If the car breaks down – I’ll fix it (costly).”

Consumer: If he’s faking it and the car is bad (potentially faked characteristic), this is very costly to him.

Result: Costly to Fake principle helped communicate credible information and secure this transaction.

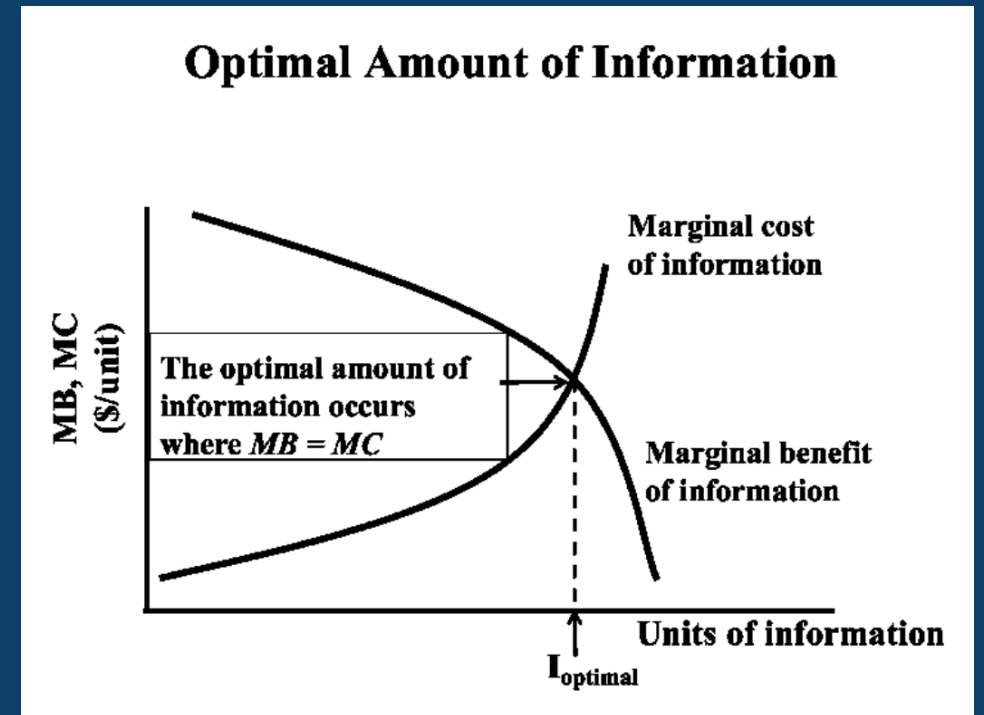
The costly-to-fake principle is the idea that a signal is believable only if it is difficult or expensive for low-quality types to imitate.

For example, a university degree can act as a signal to employers. Completing a difficult degree requires time, effort, and ability, so it is “costly” to fake. High-ability workers are more able to obtain the qualification, while low-ability workers may struggle to do so. Because the signal is costly, employers treat it as credible information about productivity.

Question 1b

Consider the optimal amount of information diagram (below) from the lecture. Which of the following would not increase the optimal length of time an individual should search for information about a product?

- i) The cost of searching decreases.
- ii) Prices for the product become more varied.
- iii) The quality of the product becomes more varied.
- iv) The product becomes more standardised (less varied).
- v) Some retailers begin offering discounts.

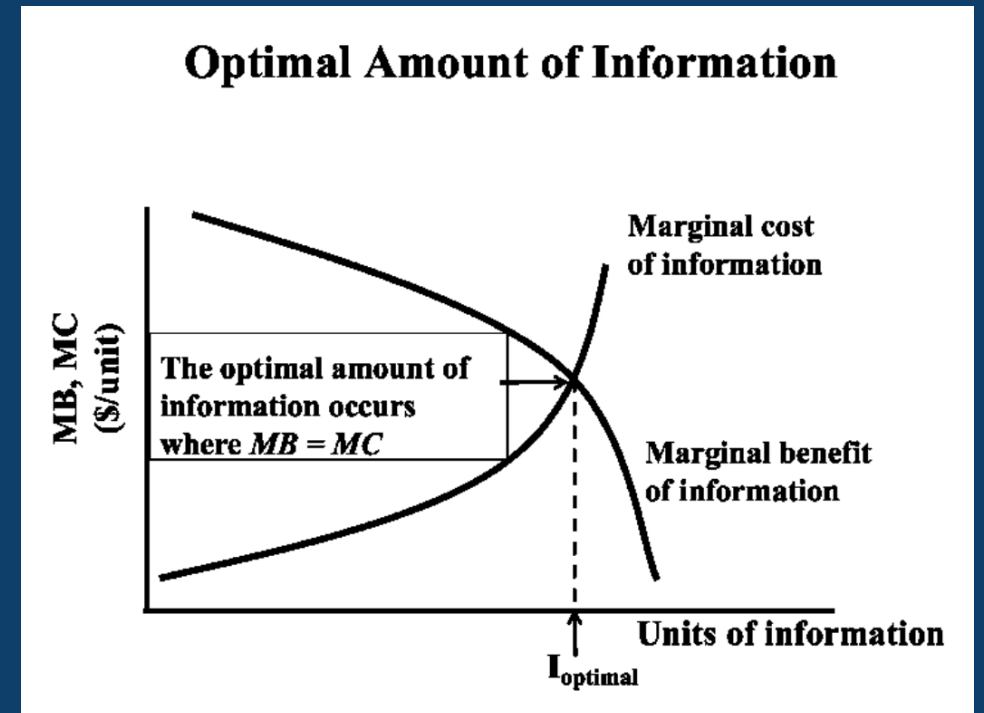


Question 1b

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- iv) The product becomes more standardised (less varied).
- v) Some retailers begin offering discounts.

A more standardised product (less variation in quality) will reduce the marginal benefit of information (leftward shift) as searching will produce more similar results – that is, the first search will be closer to the best search. When MB of information shifts left, ceteris paribus, the optimal search time (I_{optimal}) decreases.



Question 1b

i) *The cost of searching decreases.*

MC curve shifts down → crossing point moves right – more search

ii) *Prices for the product become more varied.*

MB goes up (worth searching for a better price) – more search

iii) *The quality of the product becomes more varied.*

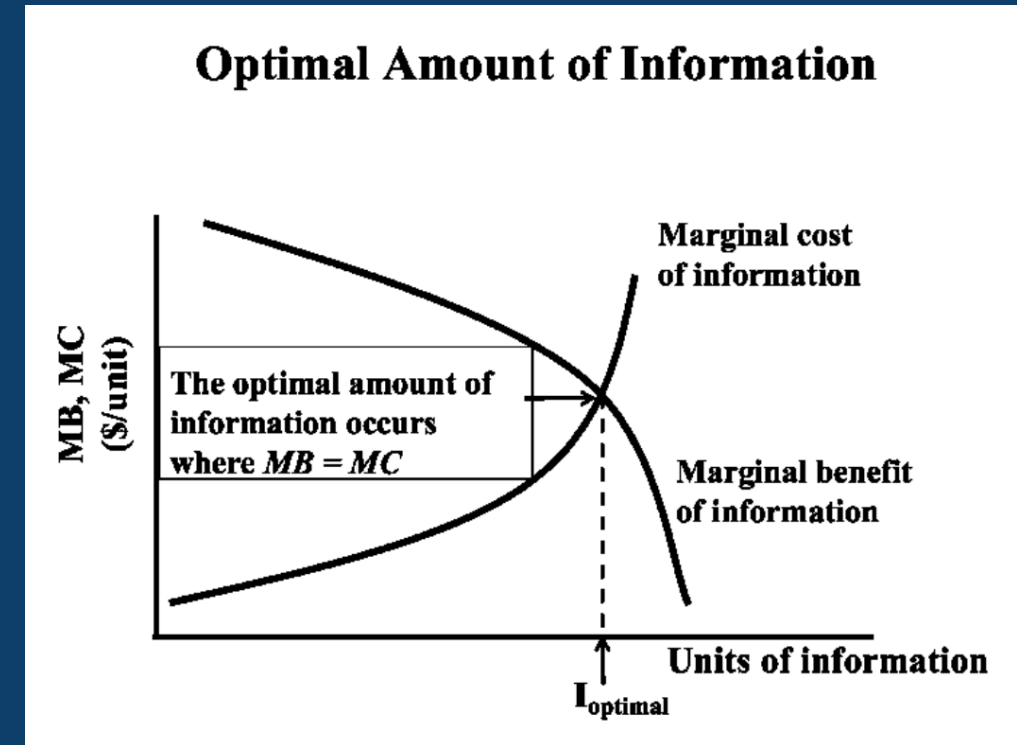
MB goes up (worth searching for better quality) – more search

iv) *The product becomes more standardised (less varied).*

MB shifts left/down (every store is similar, so why bother?) – less search

v) *Some retailers begin offering discounts.*

MB goes up (more incentive to find the deal) – more search



Question 2a

Jane is looking to rent an apartment in Melbourne's CBD. In the CBD, of the two-bedroom one-bath apartments, 75 per cent are \$800 per month and 25 per cent are \$600 per month. The marginal cost of her search rises by \$15 per search; i.e. the marginal cost of looking at the first apartment is \$15, the marginal cost of looking at the second apartment is \$30 and so on. Assuming Jane is risk neutral, the greatest number of apartments Jane is willing to look at is ____.



Question 2a

Jane is looking to rent an apartment in Melbourne's CBD. In the CBD, of the two-bedroom one-bath apartments, 75 per cent are \$800 per month and 25 per cent are \$600 per month. The marginal cost of her search rises by \$15 per search; i.e. the marginal cost of looking at the first apartment is \$15, the marginal cost of looking at the second apartment is \$30 and so on. Assuming Jane is risk neutral, the greatest number of apartments Jane is willing to look at is ____.

3, as:

MB: Expected value of MB = $(800-800) * 75\% + (800-600) * 25\% = \50 per search

MC: 1st search $MC_1 = \$15$, 2nd search $MC_2 = \$30$, 3rd search $MC_3 = \$45$, 4th search $MC_4 = \$60$

As the 3rd search $MC_3 = \$45$ and the 4th search $MC_4 = \$60$, Jane will do a maximum of 3 searches as $MC_3 = \$45$ is closest to the expected MB = \$50 per search (and in addition, $MB < MC_4$ on the 4th search.

The marginal benefit (MB) here is the expected saving from searching one more apartment. Jane wants the cheaper \$600 apartment instead of the \$800 one, so the benefit of finding a cheap apartment is: $800-600=200$

But only 25% of apartments are \$600, so the expected benefit is:

$$\text{MB} = (200 \times 0.25) + (0 \times 0.75)$$

$$\text{MB} = 50$$

$$\text{MB} = (800 - 600) \times 0.25 = 50$$

If she sees another \$800 apartment, there is no gain compared to the current expected market option: $800-800=0$ So each extra search has an expected marginal benefit of \$50.

Question 2b

Consider two types of houses: those with cracked foundations and those without. In all other respects, these two types of house are identical. Houses without cracked foundations have a known value of \$600 000, while those with cracked foundations have a known value of \$540 000 (that is, \$600 000 minus \$60 000 to fix the cracks). The frequency of cracked foundations is 20%.

Importantly, sellers know if their house foundations are cracked or not, but buyers do not know the conditions of any house foundations. All sellers will wait until they receive at least the known value of their house.

- i) How much would a risk neutral buyer be willing to pay if they are unsure if the house foundations are cracked or not?
- ii) Would any seller of a house without cracked foundations accept the purchase price in part i)?
- iii) How would your answer in ii) affect the likelihood of buyers purchasing a house with cracked foundations – increase, decrease, or have no effect? To what likelihood in the long run?
- iv) What will be the resulting market price of houses in this market in the long run?



Question 2b

i) How much would a risk neutral buyer be willing to pay if they are unsure if the house foundations are cracked or not?

A risk neutral buyer will pay the expected value of the house.

Expected value = \$600 000 * 80% + \$540 000 * 20% = \$588 000

Remember:

Risk neutral is not concerned with risk (instead focusing only on expected values) – “both are equal”.

Risk averse prefers a certain outcome over a gamble with the same expected value – “play it safe”.

Risk seeking prefers the gamble over a certain outcome “for the thrill”.

ii) Would any seller of a house without cracked foundations accept the purchase price in part i)?

No. The question says that sellers will only accept the known value. As houses without cracked foundations are known to the sellers to be worth \$600 000, they will not accept the lower price of \$588 000.

Question 2b

iii) How would your answer in ii) affect the likelihood of buyers purchasing a house with cracked foundations – increase, decrease, or have no effect? To what likelihood in the long run?

If sellers of houses without cracked foundations are not willing to accept \$588 000, only houses with cracked foundations will sell for \$588 000 as this is greater than the houses' true value (\$540 000). Therefore, the likelihood of buying a house with cracked foundations at \$588 000 increases to 100%.

iv) What will be the resulting market price of houses in this market in the long run?

When purchasers realise that the probability of buying a house with cracked foundations is 100% the expected value of houses become:

Expected value = \$600 000 * 0% + \$540 000 * 100% = \$540 000

Question 2c

In 2018 the Australian health insurance market covered about 45% of the population and received \$26.2 billion in revenues. The Australian community, however, hasn't always thought that the insurance prices are 'fair'.

- i) Explain what problems the health insurance market is likely to suffer from in general? How would these problems affect the health insurance market?
- ii) Many health insurance companies have a 12-month waiting period for pre-existing conditions. How would this impact the problems in part i)?
- iii) How would an increasingly aging population impact health insurance prices?
- iv) Would the Australian government's 10 - 30% rebate (refund) to elderly health insurance holders address the issue in part iii)?



Question 2c

i) Explain what problems the health insurance market is likely to suffer from in general? How would these problems affect the health insurance market?

Adverse Selection:

The health insurance market suffers from lack of information about their clients. Clients with pre-existing illnesses or who engage in riskier activities cost the health insurance providers more. However, as it is difficult to know each member's tendency to claim, the health insurance company must average out the costs amongst all members.

As a result, the average healthy, careful person would pay more than they'd benefit and therefore will leave or won't enter the market (MBMC).

The market will be 100% sick and risky people, while healthy careful people avoid the market. The prices will rise until $MB_{\text{sick and risky}} = MC_{\text{sick and risky}}$

Moral Hazard:

Once covered by the health insurance policy, members will be less careful. For example, a broken leg while skiing overseas may result in a \$20 000 helicopter airlift and hospital charge. If this \$20 000 was from the skier's own money, they'd be far more careful when choosing ski runs. There would be fewer broken legs and fewer \$20 000 recovery expenses. The moral hazard connected to health insurance can cause riskier behaviour than otherwise anticipated, raising health insurance prices.

Question 2c

ii) Many health insurance companies have a 12-month waiting period for pre-existing conditions. How would this impact the problems in part i)?

Paying 12-months health insurance before claiming a pre-existing injury will deter a lot of already injured / sick people from taking the insurance as the marginal cost increases substantially. This prevents the adverse selection problem where members are mostly already injured / sick. This lowers the cost for health insurance providers and allows them to offer lower prices to all members.

iii) How would an increasingly aging population impact health insurance prices?

Older people are more vulnerable to illness and injury. As more vulnerable people join health insurance policies, the average cost to cover of each member (how much the insurer must pay in hospital fees) increases. Insurers will then pass this additional cost onto the members, either specifically old age people or all ages in general. Health insurance prices will increase.

iv) Would the Australian government's 10 - 30% rebate (refund) to elderly health insurance holders address the issue in part iii)?

If insurers charge their members differently based on age – yes. The elderly, who pay more will receive a rebate (refund) to offset their increased prices.

Question 3

Consider the following situations to develop your understanding of how the Economics of Information is everywhere.

- i) Why do car insurance companies offer cheaper premiums to customers who own their vehicle outright than when the vehicle has finance owing?
- ii) Explain how introducing a no-claim bonus on fire insurance policies will affect the likelihood of moral hazard in this market?
- iii) In the real-world markets for houses and cars, why doesn't the extreme outcome implied by Q2b) eventuate?



Braver, Wiser, Kinder

- The 'academic' knowledge of this course will certainly wear off (but please, not before the exam)
- But, I believe economics can make us:
 1. **Braver** because we fear less when we understand more
 2. **Wiser** because we make better decisions when we understand the world
 3. **Kinder** because we realise people make choices for their reasons

And with that, I wish you the best for your future endeavours

Thank You
Have a good life!

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